

Orion: A Space Station Power Grid

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Abstract

The Orion Station Power Grid is an interactive browser-based simulation that demonstrates the behavior of a series electrical circuit using a space station theme. The simulation models a fusion reactor, main breaker, life support, communications, and engine systems while allowing users to adjust the source voltage, battery internal resistance, wire resistance, and subsystem resistances. It displays real-time values for current, voltage, power, efficiency, and energy losses, helping users visualize how changes in one part of the circuit affect the entire system. Although the Orion Station is fictional, the simulation is based on real electrical engineering principles used in spacecraft and other power distribution systems, making it both an engaging educational tool and a realistic representation of fundamental circuit behavior.

Introduction

Electric circuits demonstrate how voltage, current, and resistance determine the flow of electrical energy. To better understand these concepts, I first examined the Boston University Basic Circuit simulation, which models a simple circuit containing a battery, switch, resistor, and connecting wires. I also referenced *Fundamentals of Physics* by Halliday, Resnick, and Walker and *Physics for Scientists and Engineers* by Serway and Jewett, which explain the principles of Ohm's law, series circuits, electrical power, electromotive force, battery internal resistance, and voltage losses in direct-current circuits. These resources provided the theoretical foundation for the features implemented in the simulation.

While the original Boston University simulation is an excellent introduction to basic circuit behavior, I felt it could be made both more realistic and more engaging. It uses ideal wires and does not model losses due to wire or battery internal resistance. To address these limitations, I redesigned the simulation as a space station power grid and added wire resistance, battery internal resistance, real-time current and power displays, and a more interactive visual interface. Although the Orion Station is fictional, the simulation incorporates electrical concepts found in real power systems, such as voltage drops, transmission losses, and power distribution to critical subsystems. These additions provide a more realistic representation of how electrical circuits operate in engineering applications while making the simulation more engaging and educational for users.

Methods

This simulation was developed between July 8 and July 22, 2026, using several AI tools, with ChatGPT (GPT-5.5) serving as the primary development assistant. ChatGPT was used through the web interface to design the simulation, generate and revise HTML, CSS, and JavaScript code, debug issues, improve the user interface, and help write the accompanying documentation.

The development process was highly iterative. My initial prompts requested a simple interactive circuit simulation with a battery, switch, and resistors. As the project progressed, I refined the prompts to include requests such as “I like the resistance sliders. Make sure you add a second resistor that is in series with the first resistor. Whatever you pull from other sources, make sure to cite it.”

I tested the AI-generated code after each revision by running the simulation locally in a web browser. I adjusted slider values, toggled the switch, and verified that calculated quantities such as current, voltage, total resistance, power, and efficiency behaved according to Ohm's law and series circuit equations. I also checked that the user interface updated correctly and that the visual layout matched

the intended design.

Several iterations were required before reaching the final version. Early versions had disconnected wires, unrealistic layouts, incorrect resistor placement, and missing features such as wire resistance and battery internal resistance. These issues were resolved by repeatedly refining the prompts, testing the results, and requesting targeted revisions until the simulation behaved as intended.

AI was most helpful for generating the initial code structure, implementing new features, debugging programming errors, and suggesting improvements to the interface. However, human judgment was still required to decide which features to include, verify that the circuit behaved correctly, organize the layout, identify errors that AI missed, and ensure the final simulation met the project requirements.

I reviewed and tested all code and written material before submission. Although AI assisted with development, I verified the simulation's functionality and confirmed that the final version accurately represented the intended physics concepts.

Results

The simulation showed that the stability of Orion Station depends on whether enough voltage is delivered to the three subsystem resistors: Life Support, Communications, and the Engine Grid. Because all components are connected in series, the same current passes through every resistor, the connecting wires, and the reactor. However, each subsystem receives a different voltage drop because its voltage depends on its resistance. The simulation calculates the current from the reactor voltage divided by the total resistance, which includes the three subsystem resistances, wire resistance, and battery internal resistance.

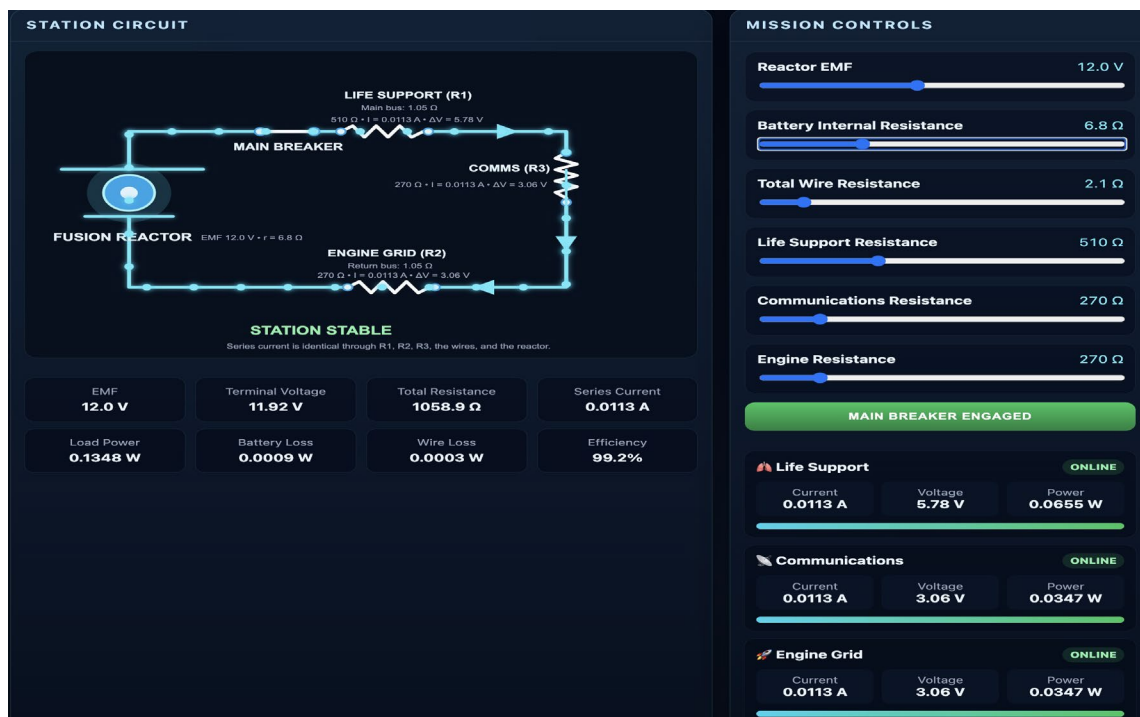


Figure 1. Example of the Orion Station Power Grid operating under stable conditions. With a reactor EMF of 12.0 V, battery internal resistance of 6.8 Ω , wire resistance of 2.1 Ω , and subsystem resistances of 510 Ω , 270 Ω , and 270 Ω , all three subsystems (Life Support, Communications, and Engine Grid) remain ONLINE. The simulation reports a series current of 0.0113 A, a terminal voltage of 11.92 V, and an overall efficiency of 99.2%, resulting in a STATION STABLE status.

In the stable configuration shown in the image above, the reactor electromotive force was 12.0 V, the battery internal resistance was 6.8 Ω , and the total wire resistance was 2.1 Ω . The subsystem resistances were 510 Ω for Life Support, 270 Ω for Communications, and 270 Ω for the Engine Grid. These settings produced a total circuit resistance of 1058.9 Ω and a series current of approximately 0.0113 A.

The program defines each subsystem's condition by comparing its voltage with a required operating

voltage. Life Support is compared with a target of 4.0 V, Communications with 3.0 V, and the Engine Grid with 3.5 V. A system is considered online when it receives at least 85% of its target voltage, limited when it receives between 55% and 85%, and critical when it receives less than 55%.

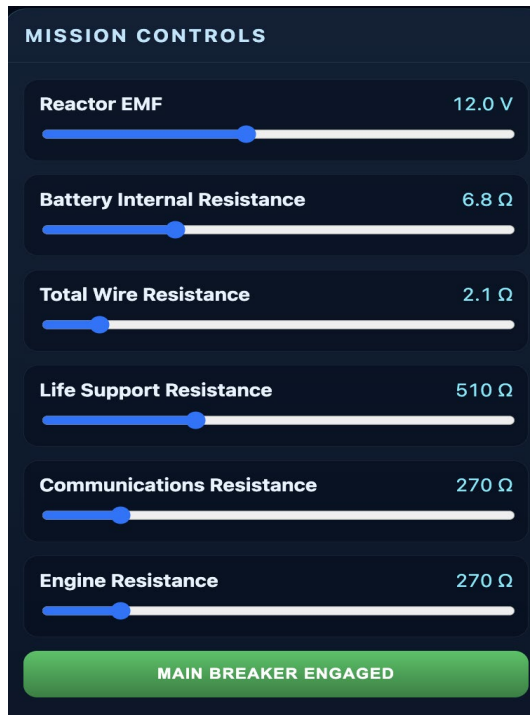


Figure 2. Mission Controls panel containing the sliders used to adjust the reactor EMF, battery internal resistance, wire resistance, and subsystem resistances during the simulation.

Using the sliders as shown above, you can adjust the following ranges of resistance. For life support, it operates online at 3.40 V or higher. It's limited from 2.20 V to below 3.40 V, and it is critical below 2.20 V. For communications, it is online at 2.55 V or higher. It is limited from 1.65 V to below 2.55 V, and it is critical when 1.65 V. For the engine grid, it becomes online at approximately 2.98 V or higher, limited from 1.65 V to below 2.55 V, and critical when below 1.65 V. The engine grid is online at approximately 2.98 V or higher, limited from approximately 1.03 V to below 2.98 V, and critical when below approximately 1.93 V. The banner displays "Station Stable" only when all three systems are online. If one or more systems become limited but none are critical, the display changes to "Power Balancing." If any subsystem becomes critical, the simulation displays "System Failure Risk."

The Life Support resistor represents equipment needed to keep the crew alive, such as oxygen circulation, carbon dioxide removal, temperature regulation, and environmental controls. In the stable trial, R1 had a resistance of 510 Ω. Because it was the largest of the three subsystem resistances, it received the largest portion of the circuit's available voltage.

The voltage across Life Support was approximately 5.78 V, which was well above its online threshold of 3.40 V. Its power consumption was approximately 0.0655 W, making it the subsystem receiving the most power in this configuration. Increasing the Life Support resistance causes it to receive a larger share of the total voltage, but it also raises the total circuit resistance and lowers the current everywhere in the circuit. This can help Life Support remain online while making Communications and the Engine Grid less stable. Decreasing the Life Support resistance causes it to receive a smaller voltage drop. Although the lower resistance increases the total circuit current, Life Support may eventually become limited or critical because too little voltage is dropped across R1.

The Communications resistor represents antennas, radios, transmitters, receivers, and other equipment used to contact mission control or nearby spacecraft. In the stable trial, the Communications resistance was 270 Ω.

The voltage across Communications was approximately 3.06 V, which was slightly above its online threshold of 2.55 V. Its power consumption was approximately 0.0347 W. Since its resistance was lower than the Life Support resistance, it received a smaller portion of the total voltage.

Reducing the Communications resistance lowers the voltage across the communications system. If it is reduced enough, the voltage will fall below 2.55 V and the system will become limited. A further reduction below 1.65 V will make it critical. Increasing its resistance raises its own share of the voltage, but the increase in total circuit resistance reduces the current through every subsystem. As a result, raising the Communications resistance too much can reduce the voltage delivered to Life Support and the Engine Grid.

The Engine Grid resistor represents electrically powered propulsion controls, navigation hardware, engine-management electronics, and power-distribution equipment associated with the spacecraft's movement. In the stable trial, the Engine Grid resistance was 270 Ω .

Its voltage was approximately 3.06 V, only slightly above its online threshold of about 2.98 V. Its power consumption was also approximately 0.0347 W. Because the Engine Grid was close to its minimum online voltage, it was the subsystem most likely to become limited after a small unfavorable adjustment.

Lowering the Engine Grid resistance reduces its voltage drop and can quickly move it below the online threshold. Increasing its resistance gives it a larger share of the source voltage, but it also reduces the series current and may decrease the voltage across the other two systems. This demonstrates why power distribution must be balanced rather than simply maximizing one subsystem's resistance.

The Reactor EMF slider controls the voltage by the station's power source. Lowering the EMF produces less current and smaller voltage drops across all three subsystems. A reduction in the EMF would first cause the engine grid to become limited in the displayed configuration because its voltage is only slightly above its online threshold. Further reductions will cause communications and eventually life support to become limited or critical. Increasing the EMF would generally improve stability by increasing current, subsystem voltage, and delivered power.

The Battery Internal Resistance slider models the electrical resistance that exists inside the fusion reactor before electrical energy reaches the rest of the circuit. Unlike an ideal voltage source, every real battery or power source has some internal resistance that causes part of the supplied voltage to be lost within the source itself. As the internal resistance is increased, a larger portion of the reactor's electromotive force (EMF) is dropped inside the battery, reducing the terminal voltage available to the external circuit. Consequently, the series current decreases, less power is delivered to the Life Support, Communications, and Engine Grid subsystems, and the battery power loss increases. This behavior demonstrates how aging batteries, damaged power systems, or inefficient power sources reduce the amount of usable energy delivered to electrical loads. In the Orion Station Power Grid simulation, increasing the Battery Internal Resistance slider makes the station progressively less stable by lowering the voltage and power available to every subsystem, while decreasing the slider represents a healthier and more efficient power source capable of supplying more energy to the station.

Finally, the wire resistance slider represents resistance in the station's power cables and buses. Increasing the wire resistance increases the total circuit resistance, causing the current to decrease. However, not all energy is delivered to the station's systems; some of it is converted into heat in the wires. Also, increasing the total wire resistance decreases the voltage across all three load resistors. In all, increasing wire resistance will make the station less stable.

Discussion

The results demonstrate that maintaining station stability requires balancing the voltage and resistance of every subsystem. Lowering the reactor voltage or increasing battery and wire resistance weakens all three systems at once. Changing an individual subsystem resistance creates a different tradeoff: decreasing it can deprive that subsystem of voltage, while increasing it can deprive the other systems of current and voltage. This models the real engineering challenge of distributing limited electrical power

among several critical loads without allowing any one system to fall below its required operating level.

One of the main strengths of the simulation is that it combines an engaging space station theme with realistic circuit behavior. Users can adjust the battery voltage, battery internal resistance, wire resistance, and individual load resistances while immediately seeing how these changes affect current, voltage, power, and efficiency. The animated electron flow and live circuit statistics make it easier to visualize the principles of a series circuit.

The main limitation is that the simulation is limited to a single series circuit with three resistive loads. It does not include more advanced components such as capacitors, inductors, LEDs, or parallel circuit branches. It also does not allow users to build or modify the circuit layout themselves.

In the future, the simulation could be extended by adding support for series and parallel circuits, additional electrical components, virtual measurement tools such as voltmeters and ammeters, and fault scenarios such as short circuits or overloaded systems. These additions would make the simulation even more realistic and educational.

Conclusion

This project improved a basic electric circuit simulation by making it more realistic and engaging. I redesigned it as the Orion Station Power Grid, where the resistors represent critical spacecraft subsystems: life support, communications, and the engine grid. Along with this new theme, I added wire resistance, battery internal resistance, real-time current displays, and improved electron animations to better model real electrical behavior. In the future, the simulation could be expanded with additional circuit components and measurement tools to create an even more comprehensive learning experience.

References